Issue	**1**
Factory ID	**25337**
Date of Issue	**10-Jul-2026**

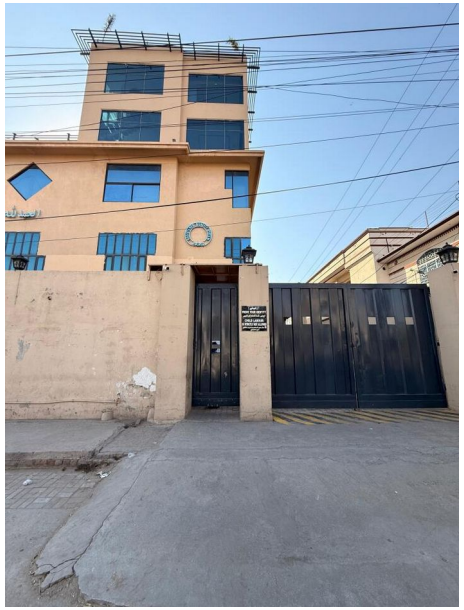
K.M Ashraf & Sons

Pull Aik, Aminabad Road
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Punjab Pakistan

Latitude: [32.478145], longitude: [74.538269]
Date of inspection: 05-May-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 05-05-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 05-May-2026, Sarah Aiman and Muhammad Bilal Butt on behalf of the Pakistan Accord carried out an initial electrical inspection for K.M Ashraf & Sons, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with Mr. Saim Khwaja (Director), Mr. Suleman Khan (Manager HR), Mr. Shujaat Hashmi (Manager Admin), Mr. Hafiz Qasim (Assistant Manager HR + Compliance) and other staff members of K. M. Ashraf & Sons. The assessment was conducted as a part of an overall Structural, Fire, and Electrical initial inspection.

The factory operates in three shifts for dyeing and security and single shift for remaining activities. Factory employs approximately 800 workers with no female representation.

The factory comprises of six main buildings and four ancillary structures owned by K. M. Ashraf & Sons. During the inspection, the buildings highlighted in blue in Figure 1.1 were surveyed with the site boundary indicated in red.

- **Building 1** consists of the following three areas:
  - **Area 1** is a five-storey (Ground + 4) building. The building is used for Penthouse (Fourth and Third Floor), penthouse (Third floor), marketing offices (Second Floor), residence, offices (First Floor) and offices (Ground Floor).
  - **Area 2** is a four-storey (Ground + 3) building. The building is used for solar panels (Roof), cutting department (Third floor), panel inspection (Second Floor), induction, ready for induction (RFI) fabric storage (First Floor) and fabric store, offices (Ground Floor).
  - **Area 3** is a five-storey (Ground + 4) building area. The building is used for solar panels (Roof), cutting (Fourth Floor) sampling (Third floor), panel inspection, embroidery (Second Floor), pad pasting (First Floor) and Fabric Inspection Department (FID); (Ground Floor).
- **Building 2** is a single-storey (Ground only). The building is used for solar panels (Roof), and stenter machines, laboratory, processing offices, jet machines, washing, chemical storage, compressor, boiler operation (Ground Floor).
- **Building 3** is a seven storey (Basement + Ground + 5) building. The building is used for solar panels (Roof), leftover store (Fifth Floor), machinery store (Fourth Floor), packing (Third & Second Floor), Audit Room (First Floor), accessory store (Ground Floor) and warehouse (Basement). The roof of the building is inaccessible.
- **Building 4** is an eight storey (Basement + Ground + 6) building. The building is used for solar panels (Roof) and stitching and finishing on all floors.
- **Building 5** is a six storey (Ground + 5) building. The building is used for bird cages (Roof), residence (Fifth, Fourth and Third Floor), canteen (Second Floor), mosque (First Floor) and offices (Ground Floor).
- **Building 6** is a single-storey (Ground Only). The building is used for solar panels (Roof), and storage (Ground Floor).
- **Ancillary Structures** consist of the following:
  - **Security Office** is a single storey (Ground Only) structure. The structure is used for office (Ground Floor).
  - **ETP Operation Room** is a single storey (Ground Only) structure. The structure is used for ETP operations monitoring.
  - **ETP Shed** is a single storey (Ground Only) structure. The structure is used to cover the ETP tanks.

- **Waste Storage Shed** is a single-storey (Ground Only structure). The structure is used for waste storage.

A summary of the main concerns in relation of life safety systems identified during the inspection (non-exhaustive) that should be actioned are as follows:

• **Abnormal Temperature**: Within electrical panels abnormal temperatures were recorded at various locations (refer section 7.2 for further details). Hotspots within panels can cause short circuits and fire due to insulation damage and overheating of cables and electrical components.

• **Fire Alarm Detection System**: Four conventional fire alarm panels were identified at the facility; however, two were found to be faulty and one was completely non-operational, leaving only a single panel in active working condition. Several issues were noted with the fire alarm and detection system, including the use of non-listed equipment, inadequate detection coverage, improper positioning of detectors, insufficient audibility, malfunctioning hooters, and fault indications on the fire alarm control panel, posing a significant life safety risk during a fire emergency.

• **Earth Missing**: In most areas, earth connection for distribution boards, electrical machines and grounding/bonding of electrical panels, electrical machines, cable trays, metal structures, metallic panel doors and metal furniture & fixings, were missing. The earthing system is important for workers operating machines, in case of short circuit/fault condition.

• **Emergency lighting System**: The emergency lighting system had inadequate coverage and was non-functional in most areas of the facility hindering safe evacuation during power outages and increasing the risk of accidents.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents

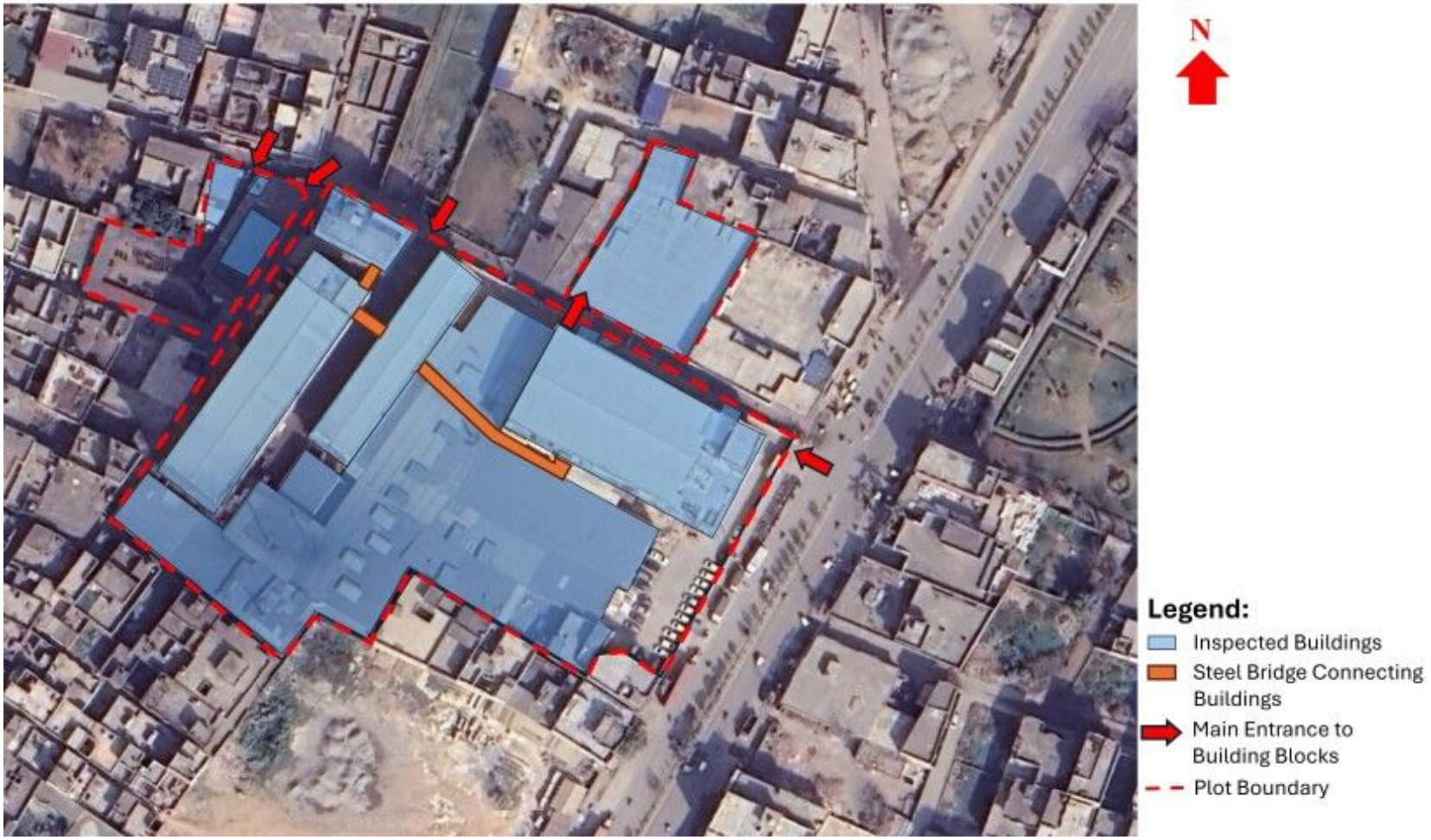


Figure 1.1. Site plan (Satellite view). Red dashed line indicates site boundary, highlighted (blue) area indicates buildings inspected.

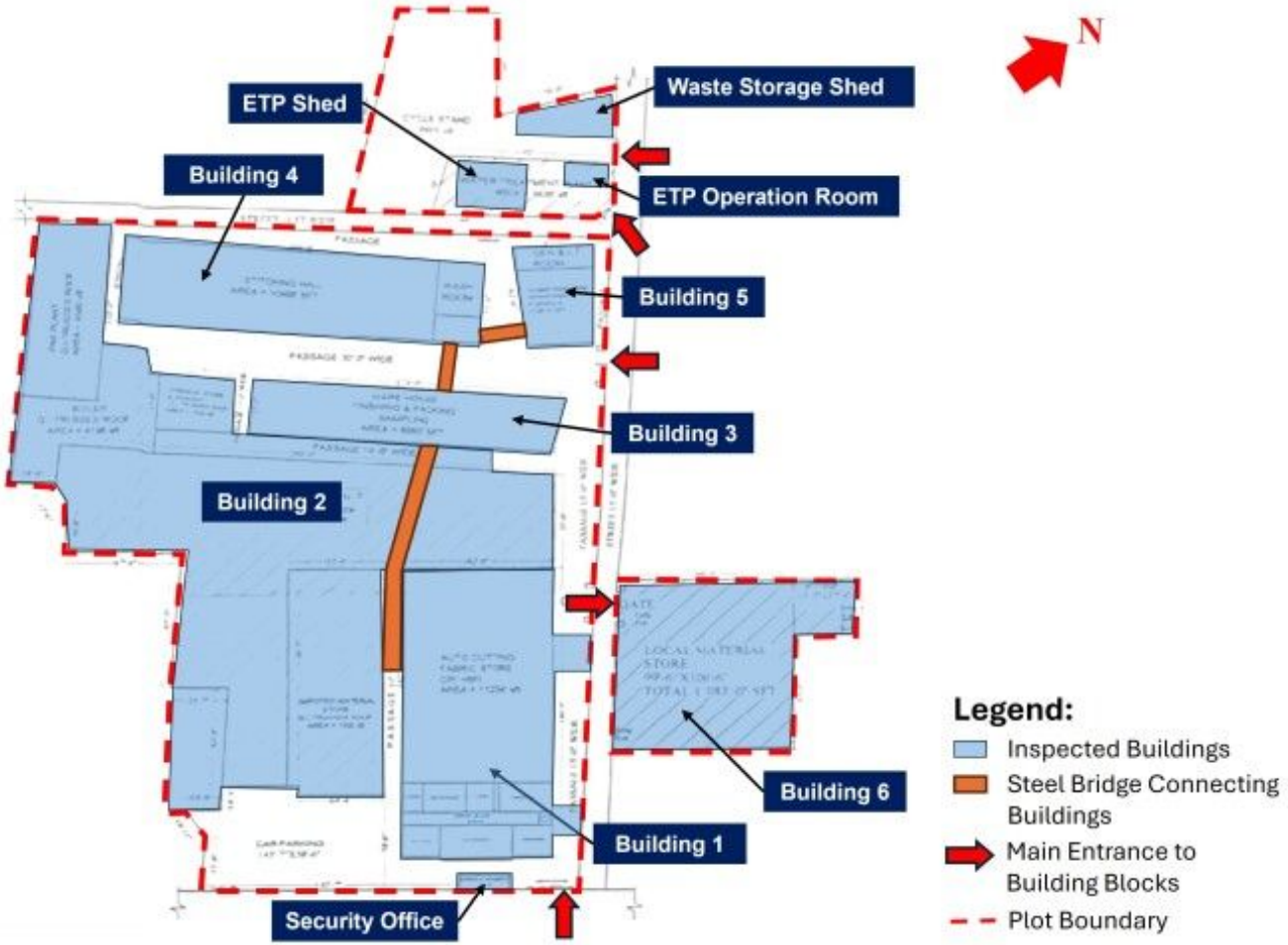


Figure 1.2. Site layout plan - Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected

2.1 Building Use

The factory comprises of six main buildings and four ancillary structures.

Building 1

Area 1
L00-Ground Floor: Offices
L01 – First Floor: Residence, offices
L02 – Second Floor: Marketing Offices
L03 –Third Floor: Penthouse
L04 – Fourth Floor: Penthouse

Area 2
L00-Ground Floor: Fabric store and offices
L01 – First Floor: Induction , ready for induction, fabric storage
L02 – Second Floor: Panel inspection
L03 – Third Floor: Cutting
L99 – Roof: Solar Panels

Area 3
L00-Ground Floor: Fabric inspectionion department
L01 – First Floor: Pad pasting
L02 – Second Floor: Panel inspection, embroidery
L03 – Third Floor: Sampling
L04 – Fourth Floor: Cutting
L99 – Roof: Solar Panels

Building 2
L00-Ground Floor: Stenter machines, laboratory, processing offices, jet machines, washing, chemical storage, compressor, boiler operation
L99 – Roof: Solar panels

Building 3
B00 - Basement : Warehouse
L00 - Ground Floor: Accessory store

L01 – First Floor: Audit room
L02 – Second Floor: Packing
L03 – Third Floor: Packing
L04 – Fourth Floor: Machinery store
L05 – Fifth Floor: Leftover store
L99 – Roof: Solar Panels

Building 4
B00 - Basement: Warehouse
L00 - Ground Floor: Stitching and finishing
L01 – First Floor: Stitching and finishing
L02 – Second Floor: Stitching and finishing
L03 – Third Floor: Stitching and finishing
L04 – Fourth Floor: Stitching and finishing
L05 – Fifth Floor: Stitching and finishing
L06 – Sixth Floor: Stitching and finishing
L99 – Roof : Solar Panels

Building 5
L00 - Ground Floor: Offices
L01 – First Floor: Mosque
L02 – Second Floor: Canteen
L03 – Third Floor: Residence
L04 – Fourth Floor: Residence
L05 – Fifth Floor: Residence
L99 – Roof : Bird cages

Building 6
L00-Ground Floor: Storage
L99 – Roof: Solar panels

Ancillary Structures
1) Security Office
L00- Ground only: Offices

2) ETP Operation Room
L00- Ground only: ETP Control room

3) ETP Shed
L00-Ground only: ETP Tank Cover

4) Waste Storage Shed
L00-Ground only: Waste Storage

2.2 Utility Source

The facility operates with two medium voltage supplies (11kV) from Gujranwala Electric Power Company, as its primary power source. The medium voltage supplies enters the facility at HT panel from where it is stepped down via transformer(1000 kVA , 400 kVA) and fed to the respective LT panels. Both of these supply are then distributed throughout the facility via overhead and underground low voltage (LV) cables.

2.3 Backup source

The facility uses three diesel generators rated at 500 kVA, 350 kVA and 550 kVA each as its backup power source. These generators are owned and maintained by the facility.

The emergency lighting system and fire alarm and detection system (FADS) have internal battery backups.

3. Documentation

This table shows the detail of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No		
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	No		

4. Electrical Safety Measures

During our electrical site inspection of K.M Ashraf & Sons, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within K.M Ashraf & Sons. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202605-066600

Timeline	Remediation Plan
Medium Term CAP-0079579	Factories should begin remediation of CAP items such as cleaning, cable organisation, and basic corrections using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202605-066602

Timeline	Remediation Plan
Medium Term CAP-0079583	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0079584	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202605-066603

Timeline	Remediation Plan
Medium Term CAP-0079585	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0079586	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202605-066601

Timeline	Remediation Plan
Medium Term CAP-0079581	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0079582	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey, the circuit breakers installed throughout the facility, inside the electrical panels and distribution boards, were inspected.

i. It was observed that the circuit breakers were installed in the electrical workshop without proper fixing supports and protective enclosures. This inadequate fixing could lead to poor connections, arcing, short circuits, or other electrical faults.

ii. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes, for example:
a. In the stitching hall 5, a 175 A circuit breaker was connected to 25 mm² PVC cables.
b. In the electrical workshop distribution board, a 50 A circuit breaker was connected to 2.5 mm² PVC cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202605-066606

Timeline	Remediation Plan
Medium Term CAP-0079600	i. Install the circuit breakers with proper fixing supports and suitable protective enclosures to ensure mechanical stability and protection from external exposure to ensure compliance with BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0079601	ii. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - Improperly fixed circuit breaker at electrical workshop



Figure 2.2 - Improperly fixed circuit breakers in electrical workshop distribution board



Figure 2.3 - 2.5 mm² cable with 50 A circuit breaker in electrical workshop



Figure 2.4 - 25mm² cable with 175 A circuit breaker at stitching hall 5

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:

i. 131 °C at Turbine control panel and 73.8 °C at Turbine main breaker
ii. 96.4 °C and 70.6 at the RO Plant near parking area
iii. 64.9°C in the distribution board of Cutting Stock Area
iv. 61.4 °C on neutral at stitching hall 3 distribution board and 51.5°C on yellow phase at stitching hall 3 distribution board

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC-insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into

contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202605-066617

Timeline	Remediation Plan
Immediate CAP-0079890	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0079891	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0079892	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating.

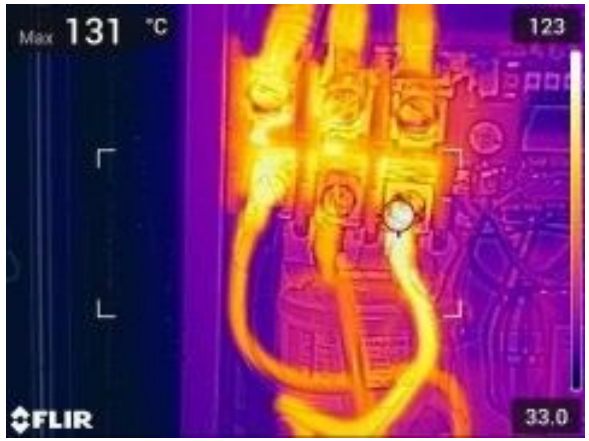


Figure 2.5 - Hotspot found at control panel of Turbine



Figure 2.6 - Hotspot found at control panel of Turbine



Figure 2.7 - Hotspot found on connection point of circuit breaker at Dyeing hall panel



Figure 2.8 - Hotspot found on connection point of circuit breaker at Dyeing hall panel



Figure 2.9 - Hotspot found on circuit breaker cable at Turbine distribution board



Figure 2.10 - Hotspot found on circuit breaker cable at Turbine distribution board

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation was provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.

ii. The earth pits were available in the facility, but it was inaccessible near LT room.

iii. The earth pit testing and maintenance records were not provided.

iv. The earthing system was almost absent in the overall facility. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. This issue was observed at stitching/production building, HR office, cutting department, embroidery, and LT panel.

v. Equipotential bonding for the cable trays and metallic panel covers was missing. This issue was observed in all areas of the facility.

vi. It was noted that the earth bar was incorrectly connected to the neutral bar in the electrical workshop distribution board, indicating improper grounding practices.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202605-066609

Timeline	Remediation Plan
Medium Term CAP-0079611	i. Design and evaluate the facility's earthing system, and develop layout drawings in accordance with BS 7671 18th Edition + A2 (2022). Submit the earthing layout drawings to the Pakistan Accord for review.
Medium Term CAP-0079612	ii. Redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.

Medium Term CAP-0079613	iii. Implement periodic earth testing per BS 7671 Regulation 643.7.1 to verify grounding effectiveness and submit testing record to Pakistan Accord for review.
Medium Term CAP-0079614	iv. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0079615	v. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Medium Term CAP-0079617	vi. Disconnect the earth bars from the neutral bars with the distribution boards and ensure proper separation and connection.
Longer Term CAP-0079616	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements. .



Figure 2.11 - Earth connection combined with neutral in electrical workshop



Figure 2.13 - Earth connection missing for the stitching machine



Figure 2.12 - Inaccessible earth pit near LT room



Figure 2.14 - Earth connection missing for the water cooler



Figure 2.15 - Equipotential bonding missing at distribution board of main accessory store



Figure 2.16 - Earth connection missing in the distribution board of cutting



Figure 2.17 - Neutral and earth cables combined in the distribution board of cutting panel inspection



Figure 2.18 - Equipotential bonding missing for cable tray at dyeing area

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. In a number of external areas, it was observed that electrical components were not suitable for the environmental conditions in which they were installed. This issue was observed at RO Plant and dyeing area.

ii. Phase separators were missing in some panels posing the risk of accidental contact between phases. This issue was systematic in most of the electrical panels and distribution boards throughout the facility.

iii. Inadequate cable dressing was observed in most of the distribution boards, with conductors not properly arranged, secured, or terminated neatly. This resulted in untidy wiring, poor segregation between circuits, and unnecessary strain on terminations, which can lead to loose connections, overheating, and difficulty during inspection or fault-finding. This issue was observed at main accessory store, sampling area, dyeing hall, cutting panel inspection, and embroidery hall.

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202605-066615

Timeline	Remediation Plan
Medium Term CAP-0079837	i. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0079880	ii. Install phase separators between adjacent live conductors of different phases within distribution boards to ensure proper segregation and protection against accidental contact, in accordance with BS 7671 Regulation 416.2.
Medium Term CAP-0079881	iii. Redress and secure cables within the distribution boards to ensure they are routed neatly and without undue stress, in compliance with BS 7671 Regulation 134.1.1 and 521.8.1, ensuring safety, accessibility, and avoidance of mechanical damage.

Longer Term CAP-0079882	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.19 - Phase seprators missing in dyeing area distribution board



Figure 2.20 - Inadequate cable dressing found in the distribution board of dyeing area



Figure 2.21 - Inadequate placement of distribution board above steam lines near compressor room



Figure 2.22 - Non-IP rated panels at RO plant area

7.5 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels. This issue was specifically observed in the distribution boards of the stitching building and cutting stock area.

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202605-066610

Timeline	Remediation Plan
Medium Term CAP-0079624	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.23 - Loop connections found inside thec distribution board of cutting stock



Figure 2.24 - Loop connections found inside distribution board at ground floor of stitching building



Figure 2.25 - Loop connections found inside distribution board at first floor of stitching building

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker. This issue was observed in the distribution board of the electrical workshop, stitching hall 3, stitching hall 1, accessory store and dyeing hall.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202605-066613

Timeline	Remediation Plan
Medium Term CAP-0079736	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).



Figure 2.26 - Multiple connections in distribution board at stitching hall 3



Figure 2.27 - Multiple connections found in the distribution board of electrical workshop



Figure 2.28 - Multiple connections found in the distribution board of accessory store



Figure 2.29 - Multiple connections found in the distribution board of dyeing hall

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were discovered and base plates were missing in the panels and distribution boards. These issues were observed at stitching hall 6, cargo lift panel at stitching building, dyeing hall and cutting department.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202605-066616

Timeline	Remediation Plan
Medium Term CAP-0079883	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0079884	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.30 - Unsealed openings inside main panel at ground floor of stitching building



Figure 2.31 - Unsealed openings observed at distribution board of stitching hall 6



Figure 2.32 - Unsealed openings observed at distribution board of dyeing area



Figure 2.33 - Unsealed openings observed at distribution board of cutting department

7.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components.

In several instances, the placement of electrical panels adjacent to machines that continuously reject heat has caused the surrounding temperature to exceed ambient levels by several degrees. This temperature rise further increases the temperature of the panels and boards nearby.

This issue was observed in compressor room, dyeing hall, and checking area in Foot ball building.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202605-066611

Timeline	Remediation Plan
Medium Term CAP-0079732	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.34 - Inadequate accessibility to panel at dyeing hall



Figure 2.35 - Inadequate accessibility to panel in compressor room



Figure 2.36 - Inadequate accessibility to panels at checking area of foot ball building

7.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels and trunking systems. This issue was observed at stitching hall 5, electrical workshop, stock cutting and RO plant.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202605-066608

Timeline	Remediation Plan
Medium Term CAP-0079604	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0079605	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.37 - Lint and dust inside distribution board at electrical workshop



Figure 2.39 - Lint and dust inside distribution board at stitching hall 5



Figure 2.38 - Lint and dust inside distribution board at cutting stock



Figure 2.40 - Lint and dust inside distribution board at RO plant

7.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was observed that no lightning protection system was installed, and there was no risk assessment provided to confirm whether it was not required.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202605-066612

Timeline	Remediation Plan
Medium Term CAP-0079733	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment as per BS EN 62305. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0079734	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant Standards such as BS EN 62305 (Protection against Lightning).

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

It was observed that four conventional fire alarm panels were installed outside the Admin Building, of which two were faulty, and one was completely non-functional. None of the panels were synchronized, resulting in a lack of coordinated alarm response across the facility. The system was found to utilize non-addressable conventional equipment, which limits precise identification of alarm source locations. Several areas lacked adequate detection coverage, and manual call points were missing at key locations throughout the facility. Visual notification devices were insufficient, non-fire-rated cables were used, and audible sounders did not meet the minimum required sound pressure levels to alert all occupants effectively. These combined deficiencies represent a serious non-compliance with fire safety standards and pose a significant life-safety risk during a fire emergency.

Listed panels & components ensure safety and performance standards. Non-listed panels may not provide reliable detection and alarm functions, which can compromise timely evacuation and firefighting efforts. Conventional fire alarm systems may lacks advanced features such as individual device identification and precise location of alarms, which can delay response times during emergencies.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202605-066614

Timeline	Remediation Plan
Immediate CAP-0079797	Ensure the functionality and performance of the existing alarm and detection system within the building.

Medium Term CAP-0079798	i. Submit design documentation for the new replacement of FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0079799	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for the import of any components required for the FADS design submitted earlier.
Longer Term CAP-0079800	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0079801	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0079802	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0079803	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.41 - Smoke detector found to be disconnected at panel inspection area



Figure 2.43 - Smoke detection missing in electrical workshop



Figure 2.42 - Non functional manual call poiny at LT room



Figure 2.44 - Smoke detection missing in director's office at Foot Ball Building



Figure 2.45 - Manual call point placed at inadequate height near generator area



Figure 2.46 - Incorrectly installed smoke detectors at dyeing area

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection, it was observed that emergency lighting was provided but not functional in a number of locations. This issue was observed at the compressor room, HR building, dyeing hall, stitching hall 6, packing , stitching hall 5, sampling hall, cutting department, and cutting stock area.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202605-066620

Timeline	Remediation Plan
Medium Term CAP-0079902	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0079903	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.47 - Non functional emergency sign at sampling



Figure 2.48 - Non-functional emergency light in the packing area



Figure 2.49 - Emergency light non-functional in the cutting stock area



Figure 2.50 - Emergency light non-functional outside HR building

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. The layouts for emergency lighting were missing. The absence of emergency lighting layouts throughout the facility raises concerns regarding the adequacy of illumination provided during power failures or emergencies.

ii. Several areas lack sufficient emergency lighting and exit directional signs coverage. This issue was specifically observed at the following locations :
a) Dyeing area
b) Generator area
c) Sampling hall
d) Stairs of cutting department
e) Pad pasting area of cutting building
f) Transformer room
g) Stitching halls

iii. External emergency lighting was not provided, particularly in the assembly area, preventing people from reaching a place of safety outside the facility during nighttime evacuations.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202605-066618

Timeline	Remediation Plan
Medium Term CAP-0079896	i. Emergency lighting layouts must be developed in accordance with the Building Code of Pakistan Fire Safety Provisions 2016 and submitted to Pakistan Accord for review.

Medium Term CAP-0079897	ii. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Medium Term CAP-0079898	iii. Install external emergency luminaires to enable safe evacuation during hours of darkness, all the way to a safe place outside the building(s).
Longer Term CAP-0079899	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.51 - External emergency lights not provided in the assembly area



Figure 2.52 - Emergency light missing in pad pasting area of cutting building



Figure 2.53 - Emergency light missing at dyeing area



Figure 2.54 - Emergency light missing in HT room

9.3 Independent power source

During the inspection, it was observed that the emergency lighting within the facility was not connected to an independent dedicated circuit.

The circuitry of emergency lights and exit signs be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202605-066619

Timeline	Remediation Plan
Longer Term CAP-0079900	Reconfigure emergency lights for independent circuit. Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Ventilation and illumination

This section refers to the transformer room, sub-station rooms and generator room.

An assessment of the existing ventilation system was conducted, alongside an evaluation of the lighting setup in various parts of the room, to determine their compliance with recommended standards.

The following observations were made:

i. The ventilation system was inadequate, with insufficient airflow leading to heat accumulation, particularly in the panel room of the dyeing area and the HT panel room and the transformer room.

ii. Several sections of the HT room and panel room of the dyeing area were poorly lit, with light levels falling below the minimum required for safe operation and maintenance in the room.

Addressing these ventilation and illumination issues is crucial for life and fire safety. Proper ventilation is essential to prevent the overheating of electrical equipment, which can lead to equipment failure or fires, posing significant risks to both personnel and infrastructure. Additionally, adequate ventilation ensures that harmful fumes or gases do not accumulate, which could be hazardous to anyone entering the room. Proper illumination is equally important as it ensures that all areas are visible, reducing the risk of accidents and enabling quick identification and response to any issues that may arise.

BS7671 18th Edition + A2 (2022) specifies the need for ventilation to prevent overheating and ensure the safe operation of electrical installations. The standard also emphasizes that adequate lighting must be provided to ensure visibility for safe operation and maintenance. Proper lighting levels are crucial to avoid accidents and to ensure that any issues with the electrical installation can be promptly identified and addressed. F-202605-066917

Timeline	Remediation Plan

Medium Term CAP-0079987	i. Hire a qualified engineer to reassess the ventilation system to ensure it meets safety and performance standards. If the assessment concludes that the current system is inadequate redesign the ventilation layout to enhance the distribution of airflow ensuring that critical areas around the HT panel and transformer receive adequate ventilation. This might involve repositioning vents or adding ductwork.
Medium Term CAP-0079988	ii. Add more lighting fixtures in poorly lit areas to ensure even distribution of light throughout the room. Focus on areas that require the highest visibility, such as around equipment and control panels.
Longer Term CAP-0079989	Establish regular maintenance and testing of the ventilation system and luminaires within the spaces to ensure adequate performance.



Figure 2.55 - Insufficient ventilation and illumination at HT panel room



Figure 2.56 - Insufficient ventilation and illumination at machine panel at dyeing area

10.2 Working clearance

This section refers to the electrical switchrooms, generator room(s), and transformer room(s).

During our inspection, the working clearances surrounding key electrical equipment were assessed.

It was observed that various items including electric fans, motors, and miscellaneous equipment were stored in the vicinity of the HT panel, significantly obstructing the required working clearance.

Proper working clearance is crucial for ensuring life and fire safety. Adequate clearance allows personnel to safely access and maintain equipment, thereby reducing the risk of accidents or injuries caused by cramped conditions. Insufficient working clearance increases the risk of equipment failure due to limited maintenance access, which can lead to fires or other hazardous situations. F-202605-066916

Timeline	Remediation Plan
Medium Term CAP-0079923	Reorganize the equipment layout to ensure adequate clearance, remove obstructions in pathways, and implement clear floor markings and signage to designate working areas and pathways.
Longer Term CAP-0079924	Conduct regular workspace audits to maintain clearances, update maintenance protocols to include clearance checks, provide personnel training on maintaining clearances, and consider installing adjustable equipment to accommodate flexible positioning.



Figure 2.57 - Insufficient working clearance around the HT panel

10.3 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

The generator’s battery terminals were found uncovered and lacking adequate insulation, leaving live metal parts directly exposed. This condition significantly increases the likelihood of accidental contact by workers, tools, or metallic objects, which could result in electrical shock.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202605-066914

Timeline	Remediation Plan
Medium Term CAP-0079919	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0079920	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.58 - Exposed terminals observed in for 500 kVA diesel generator



Figure 2.59 - Exposed terminals observed in for 550 kVA diesel generator

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection, a visual survey of the cable containment system was conducted.

Improper support for cables was observed in multiple areas. Cables were laid directly on the ground or loosely hung along walls, particularly in outside areas. This practice can cause mechanical stress and damage to the cables over time, posing safety and reliability risks. The issue was noted at generator area, outside mainthe office building and the cutting stock area.F-202605-066912

Timeline	Remediation Plan
Medium Term CAP-0079911	Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Longer Term CAP-0079912	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.60 - Inadequate support for cables outside cutting stock area



Figure 2.61 - Inadequate support for cables at generator area



Figure 2.62 - Inadequate support for cables outside main office building

11.2 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed in distribution board of electrical workshop, compressor hall , stitching hall 5, stitching hall 3, accessory store and cutting panel inspection.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was observed in distribution boards of electrical workshop, stitching hall 4 and accessory store.

iii. Low quality PVC connectors used in distribution board of fire hydrant which could be hazard of potential short circuit due to its poor insulation. This issue was observed at electrical workshop, compressor room, HR office and dyeing hall.

For any future electrical work, the termination of stranded cables must be done with cable lugs, and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety.F-202605-066913

Timeline	Remediation Plan
Medium Term CAP-0079913	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Medium Term CAP-0079914	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.

Medium Term CAP-0079915	iii. Replace the PVC connector with a standard quality terminal block in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0079916	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0079917	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0079918	iii. Establish a plan to replace all PVC connectors with compliant electrical terminal blocks and integrate this into regular maintenance to ensure ongoing safety and compliance.



Figure 2.63 - PVC tape joints and connectors observed in distribution board of electrical workshop



Figure 2.65 - Cable termination without lugs at accessory store



Figure 2.64 - PVC Connector used for connection of AC in HR office



Figure 2.66 - Cable termination without lugs at cutting panel inspection

11.3 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

This issue was observed at stitching hall 5, sttiching hall 3 and cutting panel inspection.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202605-066918

Timeline	Remediation Plan
Medium Term CAP-0079995	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0079996	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.67 - Incorrect color coding observed in the cutting panel inspection distribution board



Figure 2.68 - Incorrect colour coding observed in distribution board of stitching hall 3



Figure 2.69 - Incorrect colour coding observed in distribution board of stitching hall 5

11.4 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

Lighting tube rods and LED lights were found suspended directly from cables, and several fixtures were broken due to lack of proper support, particularly in the stitching hall 6, FID, cutting department and sampling hall.F-202605-066915

Timeline	Remediation Plan
Medium Term CAP-0079921	Install appropriate lighting fixtures with secure mounting and replace unsafe wire supports with proper systems to ensure safe and compliant installation.
Longer Term CAP-0079922	Standardize facility-wide safe lighting installation using durable supports, conduits, and proper cable management to prevent wiring stress and ensure safety.



Figure 2.70 - Light fixture on cable support in stitching hall 6



Figure 2.71 - Light fixture on cable support in sampling hall

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.